EXPLORATORY DATA ANALYSIS · 15-MINUTE TALK

Finding the needle: 492 frauds in 284,807 transactions

What the creditcard.csv dataset tells us about how fraud looks, when it happens, and whether a simple model can catch it.

Audience: technical (data science mentor) · Companion notebook: creditcard_eda.ipynb

Dataset: European cardholders, Sept 2013 (ULB / Worldline) · 2 days of transactions

ROADMAP

The story: fraud is rare, small, nocturnal, and separable

1 **Is the data clean enough?**

Quality · duplicates

2 **How imbalanced is fraud?**

Why accuracy fails

3 **Is fraud big-ticket?**

Amount distributions

4 **When does it happen?**

Hour-of-day pattern

5 **What separates it?**

V1–V28 ranking

6 **Can a simple model find it?**

PR-AUC baseline

DATA

The dataset: 284,807 transactions, 30 features, 2 days

284,807
transactions (rows)

31
columns: Time, Amount, V1–V28, Class

48 h
time span, Sept 2013

492
fraud cases (Class = 1)

- **Time** – seconds since the first transaction; **Amount** – transaction value (currency unspecified).
- **V1–V28** – PCA-transformed, anonymised; uncorrelated by construction.
- **Class** – 1 = fraud, 0 = legitimate (target).

DATA

Quality check: clean enough, with ~1k duplicate rows to flag

0
missing values

1,081
exact duplicate rows (0.38 %)

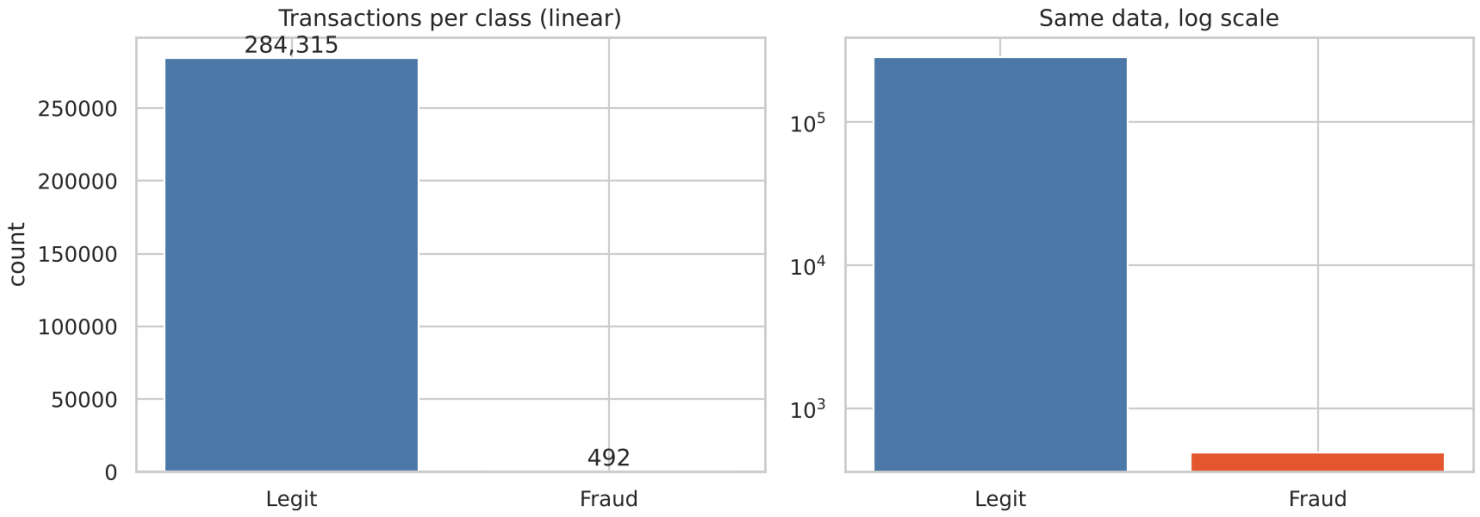
float64
all features numeric

71 MB
in memory (pandas)

- No imputation or encoding needed, so effort can go to the story.
- Duplicates kept for exploration; recommend de-duplicating before train/test split.

QUESTION 2 · IMBALANCE

Only 0.173 % of transactions are fraud – accuracy is a trap



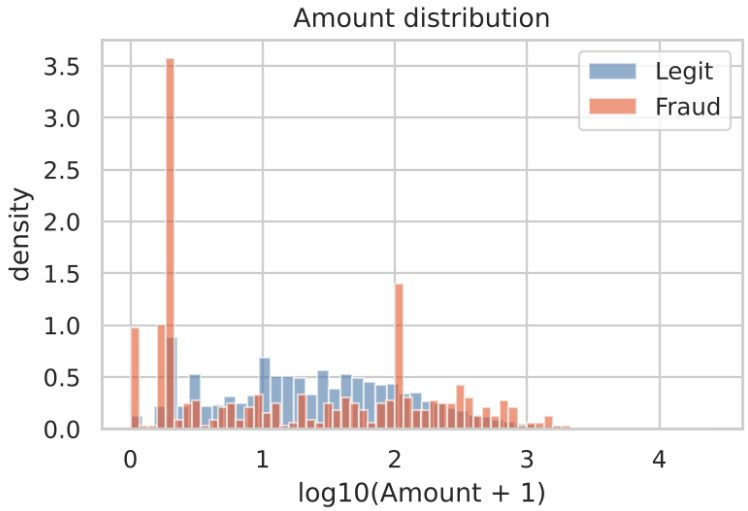
0.173 %

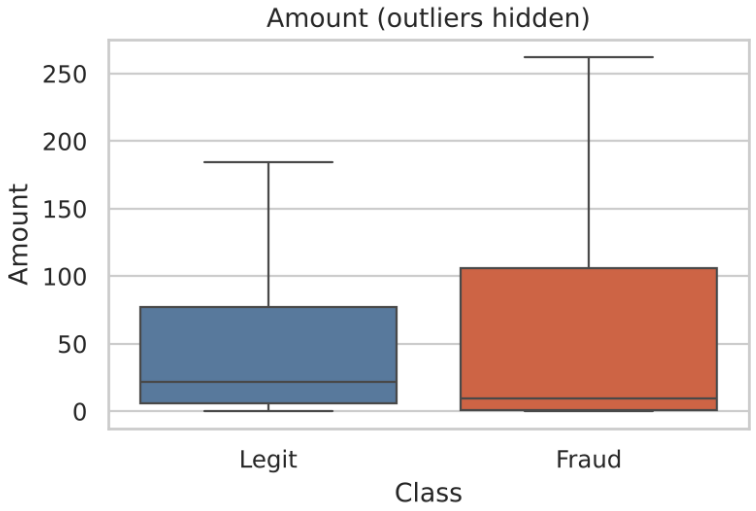
fraud rate (≈ 1 in 580)

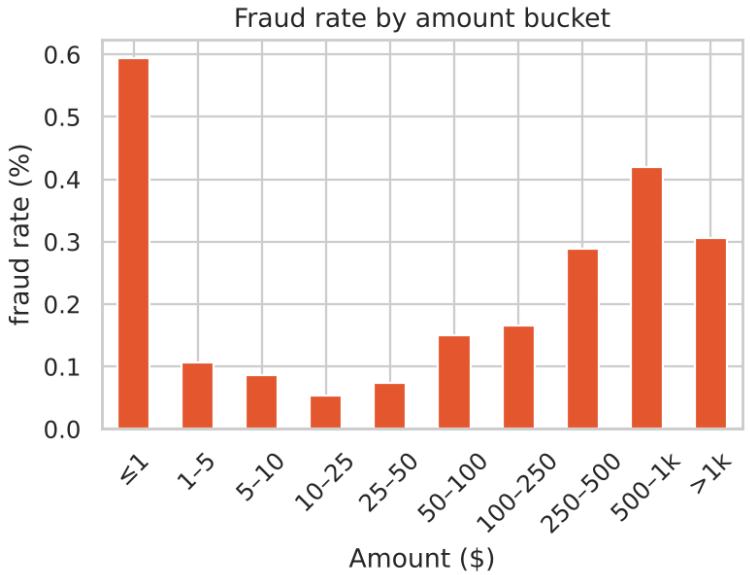
- “Always legit” = **99.83 %** accuracy, **0 %** recall.
- Evaluate with PR-AUC, precision, recall.

QUESTION 3 · AMOUNT

Fraud is small: median $9 vs $22, and 37 % of it is $1 or less



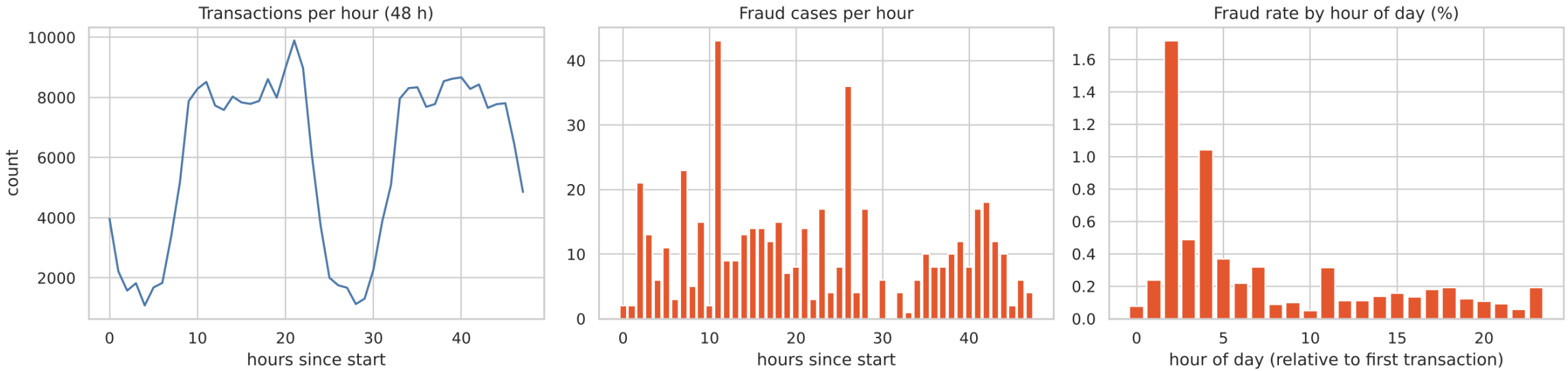




- Median: **$9.25** (fraud) vs **$22.00** (legit); max fraud only $2,126.
- **37 %** of fraud ≤ $1 vs 11 % of legit → likely card-testing.
- Amount alone is a weak signal – useful only in combination.

QUESTION 4 · TIME

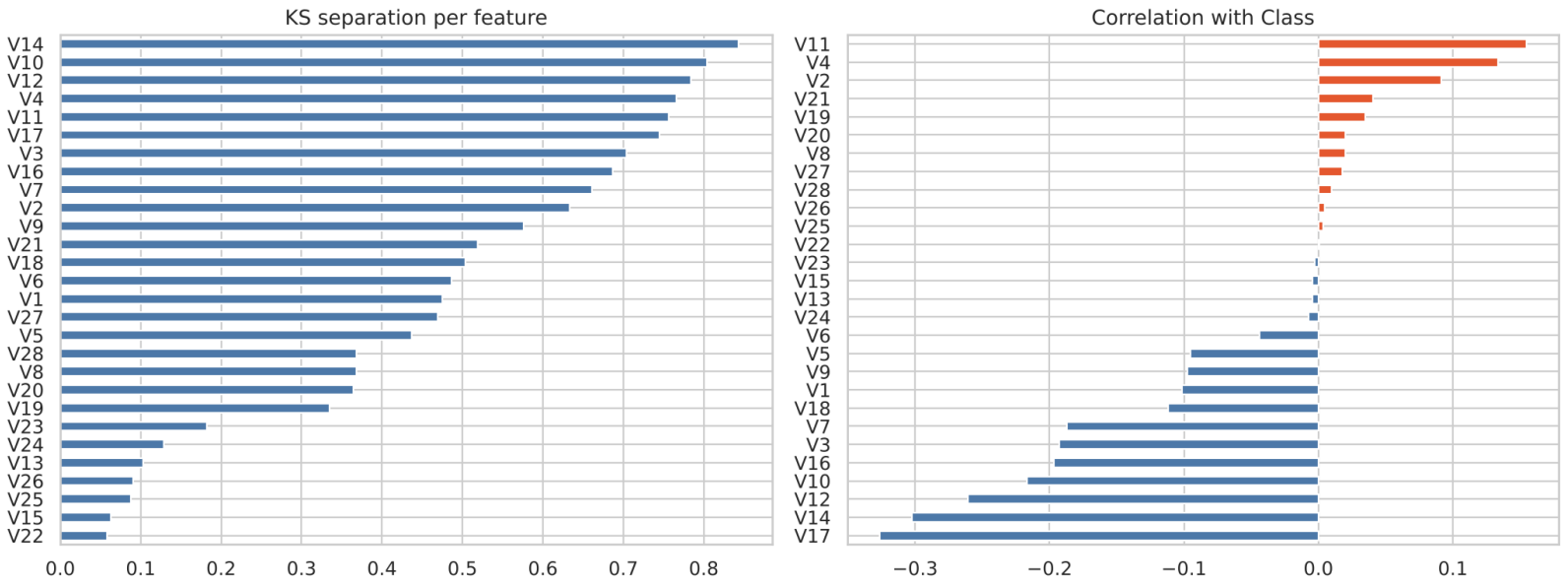
Fraud rate jumps ~5x in the quiet night hours



- Fraud counts per hour are roughly flat while traffic collapses overnight.
- Hours 1–5: **0.77 %** fraud rate vs **0.15 %** otherwise (~5x).
- Caveat: window picked after viewing the plot; only 2 days; Time is relative, not clock time.

QUESTION 5 · FEATURES

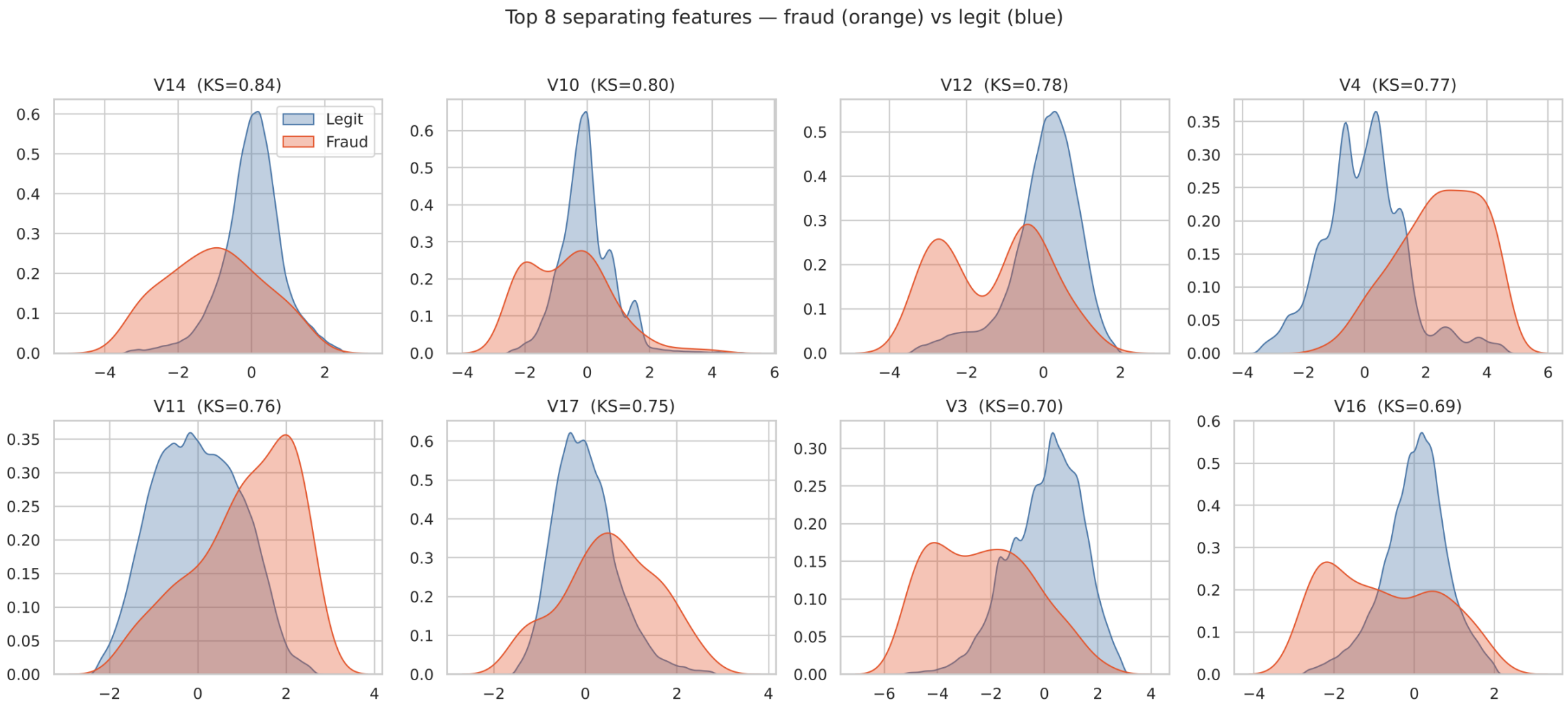
Eight anonymised components carry most of the signal



- Top KS: **V14** 0.84, **V10** 0.80, **V12** 0.78, **V4** 0.77 …
- Correlation with Class agrees: V17, V14, V12, V10 (negative); V11, V4 (positive).
- Remaining ~20 components: little signal alone.

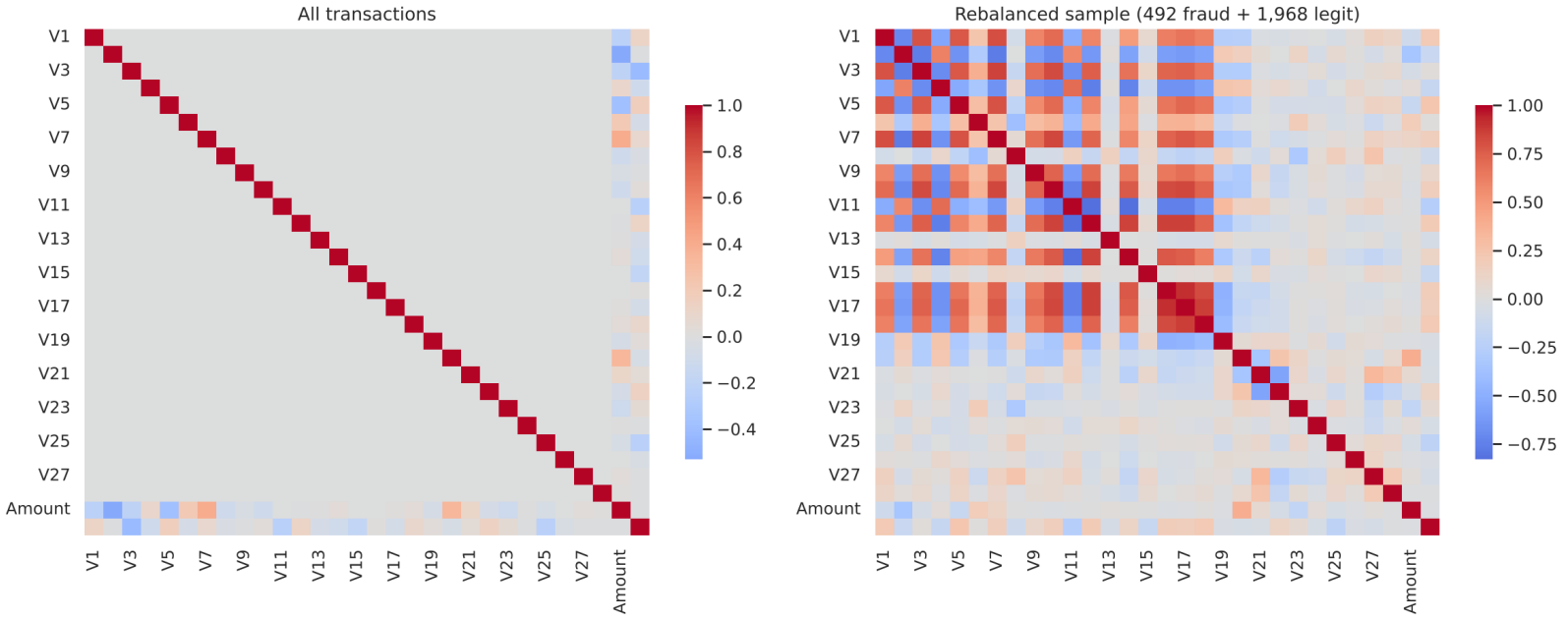
QUESTION 5 · FEATURES

Fraud sits in the tails: low V14/V12/V10/V17, high V4/V11



QUESTION 5 · REDUNDANCY

Correlations are invisible until you rebalance

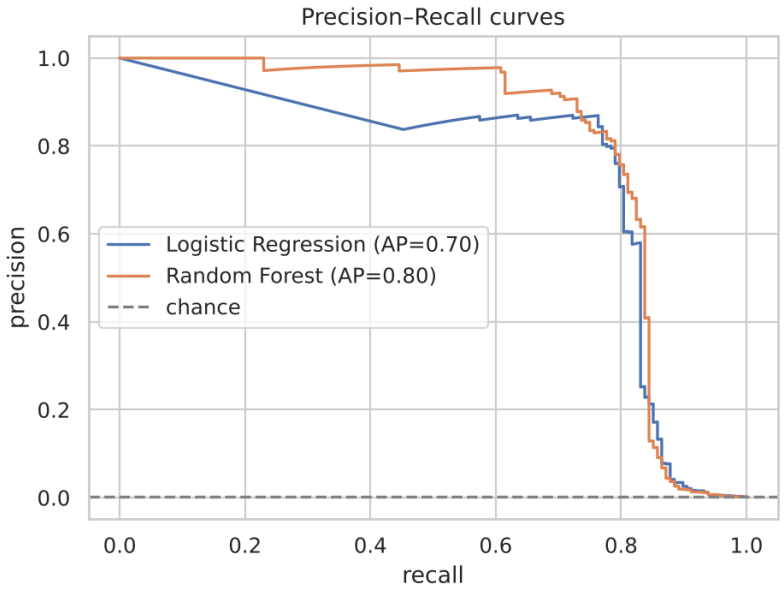


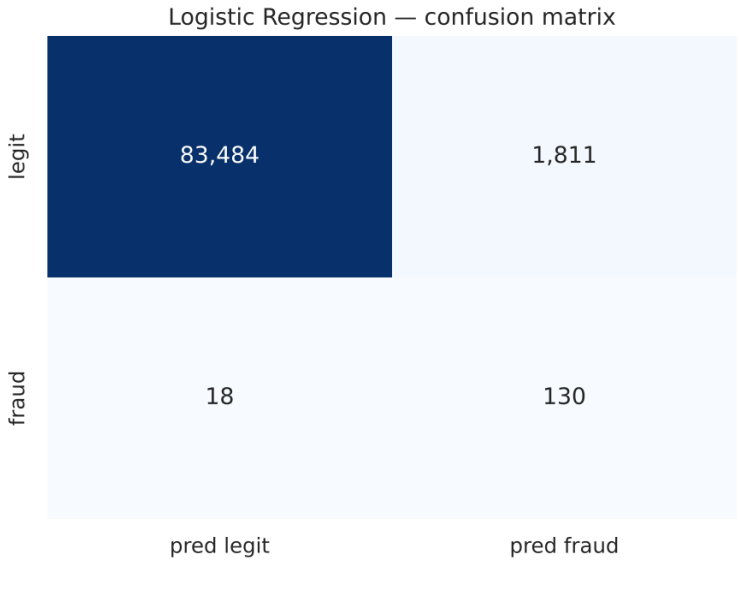
- Full data: PCA ⇒ near-zero correlation among V1–V28.
- Rebalanced sample: strong blocks (V10–V12–V14–V16–V17).
- Lesson: check correlations on a rebalanced view.

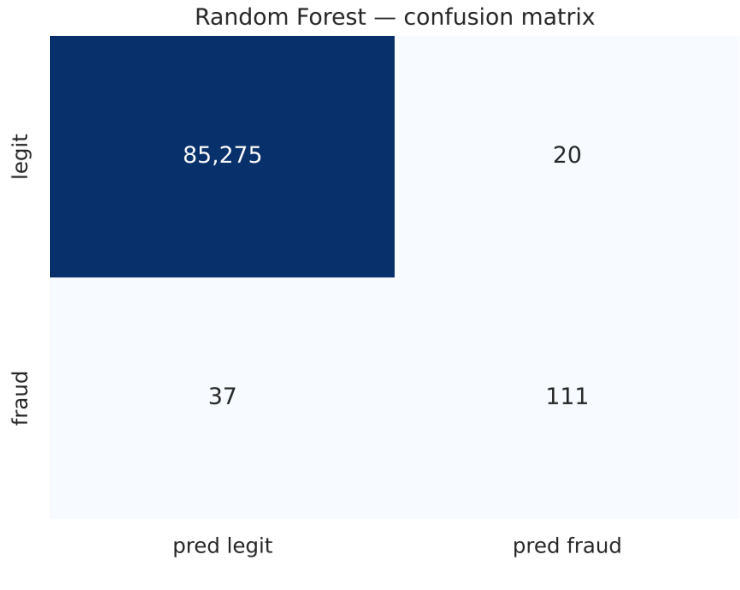
QUESTION 6 · BASELINE MODEL

A Random Forest finds 75 % of fraud at 85 % precision

Model	Accuracy	Precision	Recall	PR-AUC
Always legit	0.9983	0.000	0.000	0.002
Logistic Regression	0.9786	0.067	0.878	0.704
Random Forest	0.9993	0.847	0.750	0.803

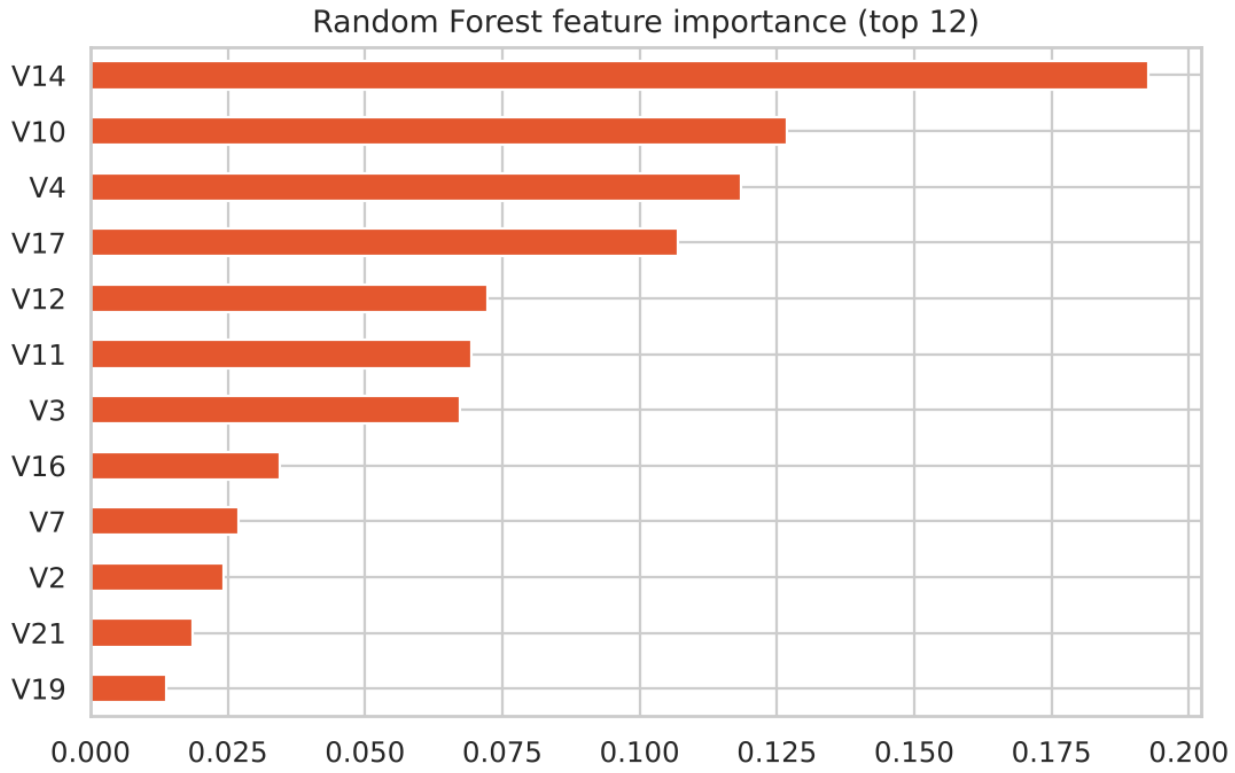






QUESTION 6 · BASELINE MODEL

Model and EDA agree: the same features matter



- Top importances: **V14** 0.19, **V10** 0.13, **V4** 0.12, **V17** 0.11, **V12** 0.07.
- Overlaps with the KS ranking: V14, V10, V12, V4, V11, V17.
- Two independent methods, one answer.

SUMMARY

Conclusions: what the data says

Question	Answer
Clean enough?	Yes: no missing values; ~1k duplicate rows to review
Imbalance?	492 / 284,807 = 0.173 % → use PR-AUC and recall, not accuracy
Amount?	Fraud is small (median ≈ $9); 37 % is ≤ $1, a card-testing pattern
Time?	Fraud rate ~5x higher in low-traffic night hours (to be confirmed)
Features?	V14, V10, V12, V4, V11, V17 carry most of the signal
Simple model?	Random Forest: PR-AUC ≈ 0.80, precision 0.85, recall 0.75

SUMMARY

Next steps and caveats

Next steps

- Tune the decision threshold against a cost matrix (missed fraud vs false alarm).
- Resampling (SMOTE / undersampling) **inside** cross-validation.
- Gradient boosting (XGBoost / LightGBM).
- Time-aware train/test split to avoid leakage.
- Anomaly-detection baselines; monitor for drift.

Caveats

- Only 2 days and 492 fraud cases.
- Features are anonymised: limited interpretability.
- Single random split; confirm with repeated / time-aware CV.
- The night-hour window was chosen after viewing the plot.